Project Report: Employee Attrition Prediction System

1. Data Understanding Report

1.1 Problem Statement

The objective of this project is to build a machine learning model that predicts whether an employee is likely to leave the organization.

Business Context

Employee attrition leads to increased recruitment costs, loss of organizational knowledge, and reduced productivity. A predictive system can:

- Reduce turnover risk
- Improve workforce planning
- Support HR in retention strategies

1.2 Objective

- Predict employee attrition (Yes/No)
- Identify key factors contributing to attrition
- Assist HR teams in proactive decision-making

1.3 Target Variable

- **Attrition**
    - 1 → Employee will leave
    - 0 → Employee will stay

1.4 Input Variables

Typical features include:

- Age
- BusinessTravel
- Department

- DistanceFromHome
- Education & EducationField
- EnvironmentSatisfaction
- Gender
- JobRole
- JobSatisfaction
- MaritalStatus
- MonthlyIncome
- OverTime
- PerformanceRating
- TotalWorkingYears
- YearsAtCompany

1.5 Initial Data Checks

Performed:

- df.shape → Dataset dimensions
- df.info() → Data types and null values
- df.describe() → Statistical summary

Observations

- Presence of categorical and numerical variables
- Possible imbalance in attrition classes (fewer employees leave than stay)

1.6 Data Cleaning Summary

- No Null values .
- No Duplicates found.
- No Error values found.

1.7 Outlier Detection & Treatment

- Technique: IQR Method

  Lower bound=Q1-1.5*IQR

  Upper bound=Q3+1.5*IQR

- Extreme outliers removed

1.8 Exploratory Analysis Summary

- **Univariate Analysis**:

  Distribution of age, income, satisfaction scores

- **Bivariate Analysis**:

  - Attrition vs OverTime,
  - Correlation between all numerical data,
  - Attrition vs JobSatisfaction,

Key Observations

- Age distribution is concentrated in mid-career professionals, which could influence attrition, promotion rates, and salary structures.
- Most employees earn below 6k per month.Employees stuck below ₹6k may feel limited and could seek opportunities elsewhere if growth isn't visible.
- while overall morale is good, there's a sizeable group of employees who may be disengaged or at risk of attrition.
- OverTime strongly impacts attrition
- Low JobSatisfaction increases attrition risk

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2. Feature Engineering Documentation

2.1 Categorical Encoding

- Label Encoding

2.2 Feature Transformation

- Log transformation applied(if skewed).
- Scaling applies where required(for linear models)

2.3 Feature Selection

- Removed non-informative features (EmployeeNumber, Over18, EmployeeCount,StandardHours)
- Checked for multicollinearity

2.4 Final Feature Set

- Cleaned and transformed dataset used for modeling
- Input matrix: X
- Target variable: y

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3. Model Development and Evaluation Report

3.1 Dataset Splitting

- Train-Test Split:
- 70% training,
- 30% testing

Output:

- X_train,X_test
- Y_train,y_test

3.2 Models Implemented

1. Decision Tree
    - Handles non-linearity
    - Easy to visualize
2. Random Forest

- Ensemble model,
- Reduces overfitting

3. Naive Bayes
    - Probabilistic baseline
4. Logistic Regression
    - Baseline model,
    - Interpretable

3.3 Model Training

- Models trained using model.fit(X_train, y_train)

3.4 Model Evaluation

Metrics used(Classification):

- Confusion Matrix
- Accuracy
- Precision
- Recall
- Misclassification Rate

3.5 Results Summary

Model	Accuracy	Precision	Recall
Logistic Regression	0.848	1.0	0.029
Naïve Bayes	0.8421	0.49	0.47
Random Forest	0.86	1.0	0.11
Decision Tree	0.79	0.33	0.29

3.6 Model Selection

- **Selected Model: Random Forest**

Reason:

- Highest accuracy and balanced precision/recall
- Handles categorical + numerical features well
- Robust to overfitting

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4. Insights and Recommendations Report

4.1 Key Insights

- OverTime is the strongest predictor of attrition
- Low JobSatisfaction and WorkLifeBalance increase attrition risk
- Income and promotion history also influence attrition
- Employees who travel frequently are more likely to leave.
- Higher salary and job level reduce attrition risk.

4.2 Feature Importance (Example)

Top contributing features:

1. OverTime
2. JobSatisfaction
3. MonthlyIncome
4. YearsAtCompany
5. WorkLifeBalance

4.3 Business Recommendations

1. **Monitor OverTime** → Reduce excessive overtime to improve retention
2. **Enhance Job Satisfaction** → Conduct regular surveys and implement engagement programs
3. **Career Development** → Provide promotion opportunities and training for younger employees
4. **Work-Life Balance** → Flexible work arrangements for high-risk groups

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4.4 Model Usage

- **Input**: Employee details
- **Output**: Attrition prediction (0/1)

4.5 Future Enhancements

- Use advanced models (XGBoost, LightGBM)
- Incorporate external HR data (performance reviews, engagement scores)
- Deploy via API for HR dashboards
- Real-time monitoring of attrition risk

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5. Conclusion

The project successfully developed a machine learning model capable of predicting employee attrition with good accuracy. The insights derived can help HR teams design proactive retention strategies, reduce turnover, and improve workforce stability.